Eswarakumar J

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Professional Summary

CSE graduate targeting Data Analytics and Machine Learning Engineer roles, with proven experience building and deploying end-to-end ML solutions across classification, NLP, and analytics. Skilled in model evaluation, model deployment, and statistical analysis; proficient in Python, SQL, and visualization tools; experienced with both classical ML and Generative AI.

Education

Government College of Engineering — Bodinayakanur, India
B.E. (Hons.) in Computer Science and Engineering; CGPA: 8.24 — Sep. 2023 – May. 2026

Alagappa Government Polytechnic College — Karaikudi, India
Diploma in Mechanical Engineering; Percentage: 88% — Jul. 2017 – Apr. 2020

Experience

Frontend Developer Intern — Jan. 2026 – Apr. 2026
Luisant Software Solutions — Salem, India

- Developed cross-platform mobile interfaces using React Native and Expo, building reusable components and navigation flows across iOS and Android.
- Handled API integration via Axios for authentication and real-time data services; utilized Expo Location and Maps APIs for live location and geospatial features.
- Followed component-driven architecture and Git-based version control workflows, collaborating within an agile team to deliver production-ready mobile screens iteratively.

Projects

Fraud Detection System | *Python, XGBoost, SMOTE, Streamlit* | Live Demo

- Architected an end-to-end binary classification pipeline on a 6.3M+ record financial dataset using SMOTE within an ImbPipeline; performed model evaluation achieving precision 0.92, recall 0.81, and F1-score 0.86.
- Engineered a real-time model deployment via Streamlit, optimizing the inference data pipeline with a decision threshold of 0.9950 through Precision-Recall curve analysis.

Semantic Book Recommendation Engine | *Python, Gemini API, ChromaDB* | Live Demo

- Built an NLP pipeline using DistilRoBERTa for multi-label emotion classification; implemented API integration with Gemini API embeddings indexed in ChromaDB for context-aware semantic search.
- Containerized with Docker and deployed an interactive Gradio dashboard on Hugging Face Spaces.

Electricity Demand Forecasting | *Python, XGBoost, Pandas*

- Processed 43,000+ hourly records over five years; engineered rolling window and time-lagged features to capture temporal dependencies.
- Trained an XGBRegressor with time-aware cross-validation and statistical analysis of residuals; applied hypothesis testing to validate model assumptions, achieving a test MAE of 110.17.

Technical Skills

Languages: Python, SQL, Javascript
Machine Learning & AI: Scikit-learn, XGBoost, Regression, Classification, NLP, Generative AI, Time-Series Forecasting, TensorFlow
Data Engineering & Analysis: Pandas, NumPy, EDA, Feature Engineering, Statistical Analysis, Hypothesis Testing, Data Pipeline, API Integration
Tools & Visualization: Tableau, MySQL, Git/GitHub, Docker, Streamlit, Jupyter

Certifications

Generative AI: Introduction and Applications | *Coursera* — 2026
IBM Data Science Professional Certificate | *Coursera* — 2025
Data Engineering Foundations | *LinkedIn Learning* — 2025
Data Science for Engineers | *NPTEL* — 2024